Day 3: Mastering Prompt Engineering for Large Language Models

Transform your AI interactions through strategic prompt design and optimization techniques

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FOUNDATION

Why Prompt Engineering Matters

Small prompt tweaks can drastically improve LLM output quality, turning vague responses into precise, actionable insights. Effective prompts reduce hallucinations and increase relevance, ensuring your AI delivers trustworthy information.

Smart prompt design unlocks the full potential of AI, transforming it from a basic tool into a powerful productivity partner that understands context and delivers exactly what you need.

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Learning Objectives

Core Concepts

Understand fundamental prompt engineering principles and how LLMs interpret instructions

Design Mastery

Master prompt design techniques for clarity, precision, and consistent results

Error Handling

Learn to detect and handle LLM hallucinations effectively

Optimization

Optimize prompts for specific use cases and achieve better outputs

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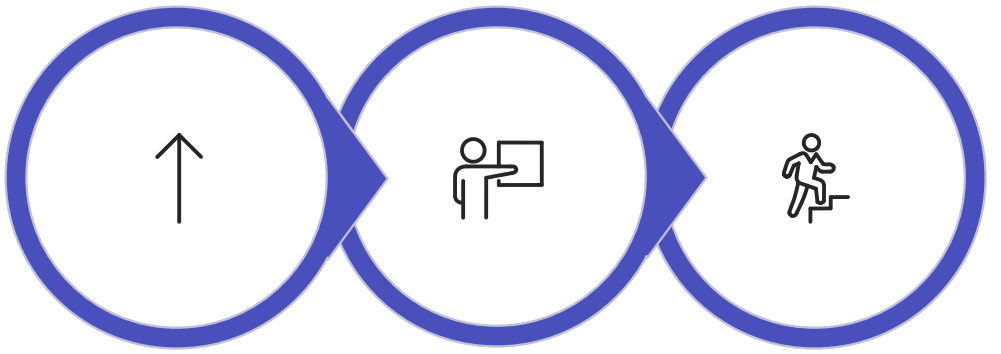
TECHNIQUES

Key Prompt Design Techniques



Zero-Shot Prompting

Instruct LLM without examples, relying solely on clear, direct instructions

Few-Shot Prompting

Provide 2-3 examples to guide model behaviour and improve output consistency

Chain-of-Thought

Encourage step-by-step reasoning for complex tasks requiring logical analysis

Universal Prompt Template Structure

```
You are a [ROLE].
Your task is to [TASK].
Context: [RELEVANT INFORMATION]
Constraints:
- Rule 1
- Rule 2
- Rule 3
Output Format: [STRICT STRUCTURE]
```

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Handling LLM Hallucinations

> **What are hallucinations?** When LLMs generate plausible but factually incorrect or fabricated information that appears convincing

Precise Prompts

Use specific instructions and constraints to narrow the model's response scope and reduce ambiguity

Retrieval-Augmented Generation

Ground responses in verified data sources and factual knowledge bases

Verification Steps

Implement fact-checking protocols and validation steps within your prompts

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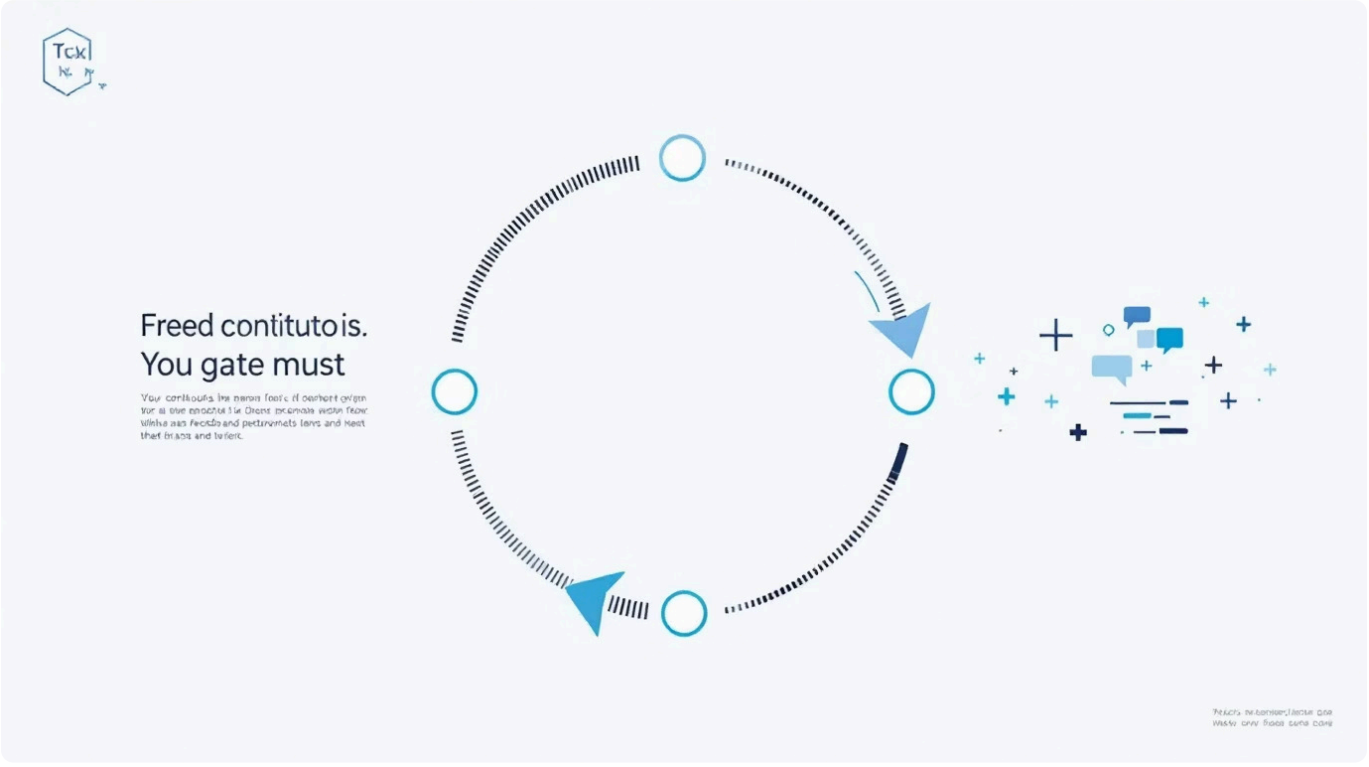
Optimizing Prompts for Better Outputs

Iterative Refinement Process

Test, evaluate, and tweak prompts systematically through multiple iterations. Each refinement cycle brings you closer to optimal results.

Use prompt templates and meta-prompts for consistency and scalability across projects. Leverage automatic prompt optimization tools and feedback loops to accelerate improvement.



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ACTIVITY 1

Study Prompt Template Patterns

01

Analyze Examples

Review effective prompt templates across different domains and use cases

02

Identify Key Elements

Spot patterns that improve clarity and reduce ambiguity in instructions

03

Discuss Adaptability

Explore how templates adapt to different tasks and domains effectively

Key Focus Areas

- Template structure and components
- Context-setting techniques
- Output formatting specifications
- Constraint definitions

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ACTIVITY 2

Practice Zero-shot and Few-shot Learning

Zero-Shot Practice

Create zero-shot prompts for simple tasks like summarization or classification

Few-Shot Design

Design few-shot prompts with 2-3 examples for complex queries requiring specific formats

Compare Results

Evaluate outputs and discuss improvements in accuracy and relevance

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ACTIVITY 3

Optimize Prompts for Specific Use Cases

Hands-On Optimization

Select a real-world use case such as customer support, coding assistance, or content generation.

Refine prompts iteratively to improve accuracy, reduce hallucinations, and enhance output quality. Test different approaches and measure results.

Share your optimized prompts with peers, discuss challenges faced, and document lessons learned for future reference.

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Your Path to Prompt Engineering Mastery

🎯 Powerful Skill

Prompt engineering harnesses AI effectively for real-world applications

📈 Continuous Growth

Practice and optimization lead to better, more reliable outputs

🔬 Experimentation

Embrace data-driven refinement for breakthrough results

Ready to build smarter prompts and unlock AI's full potential!

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